

Anup Kumar Roy

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Research Interests

- Radiation Effects and Reliability in Nanoscale CMOS Circuits
- Static Timing Analysis (STA) and Physical Design
- FPGA-Based Hardware Security Systems
- AI Hardware Acceleration and Edge Computing

Education

Lamar University, Beaumont, Texas, USA

Jan 2023 – May 2026

Master of Engineering Science in Electrical and Computer Engineering

- CGPA: 3.87/4.00
- Master's Thesis: *Impact of Temperature on Radiation-Induced Delays*
- Investigated temperature-dependent Soft Delay and Single Event Crosstalk Delay (SECD) in 22 nm CMOS technology.
- Evaluated Forward Body Bias (FBB) techniques for reliability enhancement using HSPICE simulations.

American International University-Bangladesh

May 2013 – Dec 2017

B.Sc. in Electrical and Electronics Engineering

- CGPA: 3.12/4.00

Professional Training (Ongoing)

Maven Silicon, Bangalore, India

September 2025 – Present

Job-Oriented VLSI Physical Design & Verification Training (1 Year)

- Completed core modules on **Digital Electronics** and **Verilog HDL**; currently progressing toward **ASIC Physical Design** (Floorplanning, Placement, CTS, Routing, STA, DRC/LVS).
- Training includes **EDA tools** (Synopsys, Siemens, Xilinx, Aldec), **RISC-V Core/SoC implementation**, and scripting in **Tcl & Python**.
- Focused on industry-oriented projects emphasizing **timing closure**, **PPA trade-offs**, and **physical verification practices**.

Technical Skills

EDA Tools & Flow	Vivado, HSPICE, PrimeTime, ModelSim, Cadence Virtuoso, Arduino .
Programming	Verilog, TCL, Python, C.
Documentation	L ^A T _E X, Microsoft PowerPoint.
Operating Systems	Linux, Windows.

Publications

Conferences (Selected)

- C1. R. Islam, **A. K. Roy**, A. Ahsan, M. M. Rahman Enam, M. Tanvir Islam Mim, and A. Shufian, "IoT-based Smart Municipal Garbage Management System for Fertilizer Processing System," in *2024 IEEE 3rd International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, Dhaka, Bangladesh, 2024. [Link](#) 
- C2. P Paul, **A. K. Roy**, N. Bhowmike, M. N. Islam, and X. Liu, "Design and Simulation of a Monofilar Helical Antenna for 5G and Beyond Applications," International Symposium on Measurements & Networking (M&N), IEEE, July 2024. [Link](#) 
- C3. A. Ghosh, A. S. Zishan, M. Moumita, Y. A. Kumar, **A. K. Roy**, S. Islam, S. Ahmed, H. Z. Anonto, A. T. Happy, M. S. Ahsan, H. A. Alrafai, and A. A. Hassan, "Improving the Performance of AgCdF₃-Based Perovskite Solar Cells Using Machine Learning-Driven Adjustment of Active Layer and Charge Transport Materials With SCAPS-1D," *Inorganic Chemistry Communications*,. [Link](#) 

- C4. D. Roy, M. S. Reza, A. Ghosh, **A. K. Roy**, B. R. Chowdhury, M. Aktarujjaman, H. A. Alrafai, and A. A. Hassan, “Development and Modeling of Advanced Systems Na₂SnBr₆-Based Perovskite Solar Cells: A Comprehensive Study on Electron Transport Layers,” *Optics Express*, .

Professional Experience

Maintenance Engineer

Oct 2021 – DEC 2022

HVAC systems , Dana Ltd.,Banani, Dhaka, Bangladesh.

- Maintained HVAC systems, diesel generators, UPS systems, and electrical substations.
- Performed preventive maintenance, troubleshooting, and reliability improvement activities.

Projects

A Comprehensive Digital Solution to Multipurpose AGROBOT

- Arduino IDE, Proteus, SketchUp, Bluetooth Control, Designed and developed a solar-powered multifunctional agricultural robot . [Link](#) 

Engineering Hobby Projects (YouTube)

- Developed and showcased Arduino-based embedded systems and IoT projects. [Link](#) 

References

Dr. Selahattin Sayil

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